19 Show that $\frac{8+\sqrt{12}}{5+\sqrt{3}}$ can be written in the form $\frac{a+\sqrt{3}}{b}$, where a and b are integers.

$\sqrt{12} = 2\sqrt{3}$

$(8 + 2\sqrt{3})/(5 + \sqrt{3}) \times (5 - \sqrt{3})/(5 - \sqrt{3})$

$= (40 - 8\sqrt{3} + 10\sqrt{3} - 6)/(25 - 3)$

$= (34 + 2\sqrt{3})/22$

$= (17 + \sqrt{3})/11$

1 mark
$\sqrt{12} = 2\sqrt{3}$

1 mark
multiply by conjugate

1 mark
expand numerator/denominator

1 mark
$(17+\sqrt{3})/11$

(Total for Question 19 is 4 marks)

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